

MA 587: Homework 3

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1 Question 1

1.A Weak Form

1. Start by multiplying both sides of the equation $-u'' = f$ by a test function.

$$-u''v = fv \tag{1}$$

2. Integrate both sides.

$$\int_0^1 (-u''v)dx = \int_0^1 (fv)dx \tag{2}$$

3. Integration by parts on the LHS. With the boundary conditions we are given, the first term cancels as $v(0) = v(1) = 0$.

$$\int_0^1 (-u''v)dx = -vu'|_0^1 + \int_0^1 u'v'dx = 0 + \int_0^1 u'v'dx = \int_0^1 u'v'dx \tag{3}$$

4. We then arrive at our final equation with the following expressions for the bilinear forms.

$$\int_0^1 (u'v')dx = \int_0^1 (fv)dx \tag{4}$$

$$a(u, v) = \int_0^1 (u'v')dx \tag{5}$$

$$L(v) = \int_0^1 (fv)dx \tag{6}$$

1.B Continuity and Symmetry

Symmetry of bilinear form

1. We know scalar multiplication is symmetric, so:

$$a(u, v) = \int_0^1 (u'v')dx = \int_0^1 (v'u')dx = a(v, u) \quad (7)$$

2. $a(u, v)$ is symmetric.

Continuity of bilinear form

1. Expand bilinear form.

$$a(u, v) = \int_0^1 (u'v')dx = \|u'v'\|_{L2} \quad (8)$$

2. CS to bilinear form

$$\|u'v'\|_{L2} \leq \|u'\|_{L2}\|v'\|_{L2} \quad (9)$$

3. Know the Sobolev norm is given as:

$$\|w\|_{H1} = \|w\|_{L2} + \|w'\|_{L2} \quad (10)$$

4. As each term is a norm, so each term is non-negative, we see:

$$\|w\|_{H1} \geq \|w'\|_{L2} \quad (11)$$

5. Apply to bilinear form

$$a(u, v) \leq \|u'\|_{L2}\|v'\|_{L2} \leq \|u\|_{H1}\|v\|_{H1} \quad (12)$$

6. See the bilinear form is continuous, i.e.:a

$$a(u, v) \leq \|u\|_{H1}\|v\|_{H1} \quad (13)$$

Continuity of linear form

1. Take absolute value of linear form and apply CS

$$|L(v)| = \left| \int dx f v \right| = \|fv\|_{L2} \leq \|f\|_{L2}\|v\|_{L2} \quad (14)$$

2. Similar to the continuity of the bilinear form:

$$\|w\|_{H1} = \|w\|_{L2} + \|w'\|_{L2} \quad (15)$$

3. As each term is a norm, so each term is non-negative, we see:

$$\|w\|_{H1} \geq \|w\|_{L2} \quad (16)$$

4. See the linear form is continuous as f is square integrable over the domain, so is a fixed constant

$$|L(v)| \leq \|f\|_{L2} \|v\|_{L2} \leq \|f\|_{H1} \|v\|_{H1} \quad (17)$$

$$|L(v)| \leq \|f\|_{H1} \|v\|_{H1} = C \|v\|_{H1} \quad (18)$$

1.C Coersivity of bilinear form

1. Start with the following

$$a(v, v) = \int dx v' v' = \|v'\|_{L2}^2 \quad (19)$$

2. From Poincare inequality,

$$\|v\|_{L2}^2 \leq \|v'\|_{L2}^2 \quad (20)$$

3. Sobolev norm

$$\|v\|_{H1}^2 = \|v\|_{L2}^2 + \|v'\|_{L2}^2 \leq \|v'\|_{L2}^2 + \|v'\|_{L2}^2 = 2\|v'\|_{L2}^2 = 2a(v, v) \quad (21)$$

4. See the linear form is coersive with $C = 2$

$$\|v\|_{H1}^2 \leq 2a(v, v) \quad (22)$$

1.D FEniCSx

Gemini Prompt Hello robot, I am doing a finite element method homework and need your help writing code using the FEniCSx framework. Here is the problem statement. I do not need you to answer or explain parts a-c. Only write a python code for part d.

1. Make sure the code creates a graph of the results using matplotlib

2. Call "domain.topology.create_connectivity" before trying to find the boundary facets
3. I am using a newer version of FEniCSx (v0.7 or newer)

Results [!h]

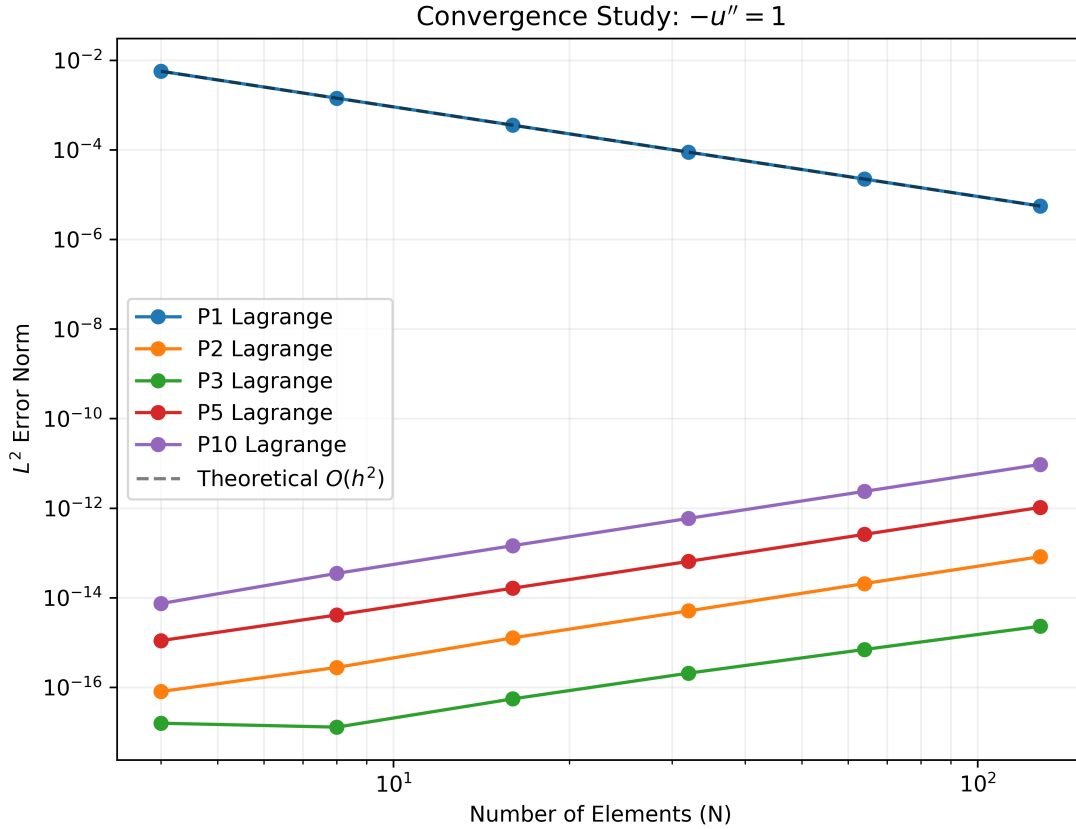


Figure 1: Convergency Study: varying polynomial degree and element county

We see the P1 Lagrange decreases in accordance with the expected convergence rate as element count increases. However, the higher degree polynomials increase with increasing element count because the polynomials are needlessly high order for such a simple function, $f(x) = 1$. Instead of the higher degree polynomials get to the machine error and quickly begin accumulating numerical error from the imperfect solver.

2 Poincare Inequality

2.A Extending 1c

1. Start with the following and apply boundary conditions to cancel $v(0, y)$.

$$v(x, y) = v(0, y) + \int_0^x \frac{\partial v(x', y)}{\partial x} dx' = \int_0^x \frac{\partial v(x', y)}{\partial x} dx' \quad (23)$$

2. Apply CS

$$v^2(x, y) = \left[\int_0^x \frac{\partial v(x', y)}{\partial x} dx' \right]^2 \leq \left[\int_0^x 1^2 dx' \right] \left[\int_0^x \left(\frac{\partial v(x', y)}{\partial x} \right)^2 dx' \right] \quad (24)$$

3. Know the max value of $x = 1$, so that is the upper bound on v^2

$$v^2(x, y) \leq [1] \left[\int_0^1 \left(\frac{\partial v(x', y)}{\partial x} \right)^2 dx' \right] \quad (25)$$

4. Integrate both sides over Ω

$$\int_{\Omega} v^2 d\mu \leq \int_0^1 dx \int_0^1 dy \int_0^1 dx' \left(\frac{\partial v(x', y)}{\partial x} \right)^2 \quad (26)$$

$$\int_{\Omega} v^2 \leq \int_0^1 dy \int_0^1 dx' \left(\frac{\partial v(x', y)}{\partial x} \right)^2 = \int_{\Omega} d\mu \left(\frac{\partial v}{\partial x} \right)^2 \quad (27)$$

5. Know the following

$$\|\nabla v\|^2 = \left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \quad (28)$$

6. Also know $v_y^2 \geq 0$, so

$$\left(\frac{\partial v}{\partial x} \right)^2 \leq \|\nabla v\|^2 \quad (29)$$

7. Arrive at:

$$\int_{\Omega} v^2 d\mu \leq \int_{\Omega} d\mu \left(\frac{\partial v}{\partial x} \right)^2 \leq \int_{\Omega} d\mu \|\nabla v\|^2 \quad (30)$$

$$\int_{\Omega} v^2 d\mu \leq \int_{\Omega} d\mu \|\nabla v\|^2 \quad (31)$$

2.B Boundary conditions on Gamma

1. Know the Sobolev norm is given as where the multi-index sum is really two sums, but I am lazy and don't want to write it all out:

$$\|v\|_{H^2}^2 = \|v\|_{L^2}^2 + \|\nabla v\|_{L^2}^2 + \sum_{i,j=1}^2 \left\| \frac{\partial^2 v}{\partial x_i \partial y_j} \right\|_{L^2}^2 \quad (32)$$

2. Expand Laplacian

$$|\Delta v|^2 = \left(\frac{\partial v}{\partial x_1} \right)^2 + \left(\frac{\partial v}{\partial x_2} \right)^2 + 2 \frac{\partial v}{\partial x_1} \frac{\partial v}{\partial x_2} \quad (33)$$

3. Integrate over Ω

$$\int_{\Omega} |\Delta v|^2 = \int_{\Omega} d\mu \left[\left(\frac{\partial v}{\partial x_1} \right)^2 + \left(\frac{\partial v}{\partial x_2} \right)^2 + 2 \frac{\partial^2 v}{\partial x_1 \partial x_2} \right] \quad (34)$$

4. We see the RHS integrand is equal to

$$\left(\frac{\partial v}{\partial x_1} \right)^2 + \left(\frac{\partial v}{\partial x_2} \right)^2 + 2 \frac{\partial^2 v}{\partial x_1 \partial x_2} = \sum_{i,j=1}^2 \left\| \frac{\partial^2 v}{\partial x_i \partial y_j} \right\|_{L^2}^2 = |\Delta v|^2 \quad (35)$$

5. From part a, we see:

$$\|v\|_{L^2}^2 \leq C_1 \|\nabla v\|_{L^2}^2 \quad (36)$$

6. We can recursively apply the above relation to the first equation in this part to obtain the following.

We see the general case is the L2 norm of the derivative of a function bounds the L2 norm of a function on Ω .

$$\|v\|_{H^2}^2 = \|v\|_{L^2}^2 + \|\nabla v\|_{L^2}^2 + |\Delta v|_{L^2}^2 \leq C_1 \|\nabla v\|_{L^2}^2 + |\Delta v|_{L^2}^2 \leq C |\Delta v|_{L^2}^2 \quad (37)$$

7. Arrive at the following

$$\|v\|_{H^2}^2 \leq C \int_{\Omega} |\Delta v|^2 d\mu \quad (38)$$

3 Question 3

3.A Weak form

1. Start with the following

$$-\Delta u = f \Rightarrow -\Delta uv = fv \Rightarrow -\int_{\Omega} d\mu \Delta uv = \int_{\Omega} d\mu fv \quad (39)$$

2. Apply GDT to the LHS

$$-\int_{\Omega} d\mu \Delta uv = \int_{\Omega} d\mu \nabla v \nabla u - \int_{\Gamma} dS \frac{\partial u}{\partial n} v \quad (40)$$

3. Use the second equation we are given

$$\frac{\partial u}{\partial n} = g - \gamma u \quad (41)$$

4. Substitute the previous two relations in the first equation

$$\int_{\Omega} d\mu \nabla v \nabla u - \int_{\Gamma} dS (g - \gamma u) v = \int_{\Omega} d\mu fv \quad (42)$$

5. Separate linear and bilinear terms

$$\int_{\Omega} d\mu \nabla v \nabla u + \int_{\Gamma} dS \gamma uv = \int_{\Omega} d\mu fv + \int_{\Gamma} dS gv \quad (43)$$

6. We obtain the following bilinear and linear forms

$$a(u, v) = \int_{\Omega} d\mu \nabla v \nabla u + \int_{\Gamma} dS \gamma uv \quad (44)$$

$$L(v) = \int_{\Omega} d\mu fv + \int_{\Gamma} dS gv \quad (45)$$

3.B Symmetry and Continuity

Symmetry of bilinear form

1. We see each component of the bilinear form is symmetric as multiplication is commutative such that:

$$\int_{\Omega} d\mu \nabla v \nabla u = \int_{\Omega} d\mu \nabla u \nabla v \quad (46)$$

$$\int_{\Gamma} dS \gamma uv = \int_{\Gamma} dS \gamma vu \quad (47)$$

2. Therefore, the bilinear form is symmetric

Continuity of bilinear form

1. Start with CS on the Ω term

$$\left| \int_{\Omega} d\mu \nabla u \nabla v \right| \leq \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} \quad (48)$$

2. With the Sobolev norm:

$$\|\nabla u\|_{H^1}^2 = \|u\|_{L^2(\Omega)}^2 + \|\nabla u\|_{L^2(\Omega)}^2 \quad (49)$$

3. We see

$$\|\nabla u\|_{L^2} \leq \|\nabla u\|_{H^1}^2 \quad (50)$$

4. Similarly for v

$$\|\nabla v\|_{L^2} \leq \|\nabla v\|_{H^1}^2 \quad (51)$$

5. Put the two relations together:

$$\left| \int_{\Omega} d\mu \nabla u \nabla v \right| \leq \|\nabla u\|_{H^1}^2 \|\nabla v\|_{H^1}^2 \quad (52)$$

6. Next, CS on the Γ term:

$$\left| \int_{\Gamma} dS \gamma uv \right| \leq |\gamma| \|u\|_{L^2(\Gamma)} \|v\|_{L^2(\Gamma)} \leq |\gamma| \|u\|_{H^1(\Gamma)} \|v\|_{H^1(\Gamma)} \quad (53)$$

7. Expand the Sobolev norm over gamma

$$\|u\|_{H^1(\Gamma)}^2 = \int_{\Gamma} dS u^2 + \int_{\Gamma} dS \left| \frac{\partial u}{\partial s} \right|^2 = \|u\|_{L^2(\Gamma)}^2 + \|\nabla u\|_{L^2(\Gamma)}^2 \quad (54)$$

8. We previously found the lower dimensional version of the following:

$$\|u\|_{L^2(\Gamma)}^2 \leq \|\nabla u\|_{L^2(\Gamma)}^2 \quad (55)$$

9. Apply the above relations:

$$\|u\|_{H^1(\Gamma)}^2 = \|u\|_{L^2(\Gamma)}^2 + \|\nabla u\|_{L^2(\Gamma)}^2 \leq 2\|\nabla u\|_{L^2(\Gamma)}^2 \quad (56)$$

10. We then use GDT to express:

$$\int_{\Gamma} dS \left| \frac{\partial u}{\partial s} \right|^2 = \int_{\Omega} d\mu |\Delta u|^2 = \|\Delta u\|_{L^2(\Omega)}^2 \leq \|u\|_{H^1(\Omega)}^2 \quad (57)$$

11. We then see:

$$\|u\|_{H^1(\Gamma)}^2 \leq 2\|u\|_{H^1(\Omega)}^2 \quad (58)$$

12. Substitute back in:

$$\left| \int_{\Gamma} dS \gamma uv \right| \leq 2|\gamma| \|u\|_{L^2(\Omega)}^2 \|v\|_{L^2(\Omega)}^2 \leq C_1 \|u\|_{H^1(\Omega)}^2 \|v\|_{H^1(\Omega)}^2 \quad (59)$$

13. Combine the two inequalities:

$$|a(u, v)| = \left| \int_{\Omega} d\mu \nabla u \nabla v + \int_{\Gamma} dS \gamma uv \right| \leq \|u\|_{H^1(\Omega)}^2 \|v\|_{H^1(\Omega)}^2 + C_1 \|u\|_{H^1(\Omega)}^2 \|v\|_{H^1(\Omega)}^2 \quad (60)$$

$$|a(u, v)| \leq C \|u\|_{H^1(\Omega)}^2 \|v\|_{H^1(\Omega)}^2 \quad (61)$$

Continuity of linear form

1. Start with CS on the integral over Ω

$$\int_{\Omega} d\mu f v \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \quad (62)$$

2. Next, the same for the integral over Γ

$$\int_{\Gamma} dS g v \leq \|g\|_{L^2(\Gamma)} \|v\|_{L^2(\Gamma)} \quad (63)$$

3. Use relationship from continuity of bilinear form

$$\|g\|_{L^2(\Gamma)} \|v\|_{L^2(\Gamma)} \leq \|g\|_{L^2(\Gamma)} \|v\|_{H^1(\Gamma)} \leq 2\|g\|_{L^2(\Gamma)} \|v\|_{L^2(\Omega)} \quad (64)$$

4. Combine the two:

$$|L(v)| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + 2\|g\|_{L^2(\Gamma)} \|v\|_{L^2(\Omega)} = (\|f\|_{L^2(\Omega)} + 2\|g\|_{L^2(\Gamma)}) \|v\|_{L^2(\Omega)} \quad (65)$$

5. With the assumption f, g are square integrable over Ω, Γ , respectively, we know the quantity in the () is a constant

$$|L(v)| \leq C\|v\|_{L^2(\Omega)} \leq C\|v\|_{H^1(\Omega)} \quad (66)$$

6. See linear form is continuous

$$|L(v)| \leq C\|v\|_{H^1(\Omega)} \quad (67)$$

3.C Coersivity

1. Start by expanding both sides of the coersivity inequality

$$a(v, v) = \int_{\Omega} d\mu |\nabla v|^2 + \gamma \int_{\Gamma} dS v^2 \quad (68)$$

$$\alpha \|v\|_{H^1(\Omega)}^2 = \alpha \left[\int_{\Omega} d\mu v^2 + \int_{\Omega} d\mu |\nabla v|^2 \right] \quad (69)$$

2. Write the expanded coersivity expression

$$\int_{\Omega} d\mu |\nabla v|^2 + \gamma \int_{\Gamma} dS v^2 \geq \alpha \left[\int_{\Omega} d\mu v^2 + \int_{\Omega} d\mu |\nabla v|^2 \right] \quad (70)$$

$$\gamma \int_{\Gamma} dS v^2 \geq \alpha \int_{\Omega} d\mu v^2 + (\alpha - 1) \int_{\Omega} d\mu |\nabla v|^2 \quad (71)$$

3. We see all integrals will evaluate to a positive quantity as the integrand is some quantity squared. We also know $\alpha > 0$.

- (a) When $\gamma > 0$, the bilinear form is coersive because we the first two terms will be positive and we can choose some positive α small enough such that $(1 - \alpha) < 0$ and the RHS will be negative
- (b) When $\gamma = 0$, $0 \geq RHS$, so the only way this expression could hold is if $v, \nabla v = 0$, which is a trivial solution. Therefore, the bilinear form is not coersive
- (c) When $\gamma < 0$, (negative quantity) $\geq \alpha$ (positive quantity) + $(1 - \alpha)$ (positive quantity). This could only be true when $v = 0$, which is a trivial case again. Therefore, the bilinear form is not coersive

3.D FEniCSx

Gemini Prompt [!h] Thank you. Next, can you write the code for part d of this problem? Follow the same instructions about plotting and versioning as you did for the previous problem. Also, print the errors formatted in a LaTeX table that I can simply copy and paste.

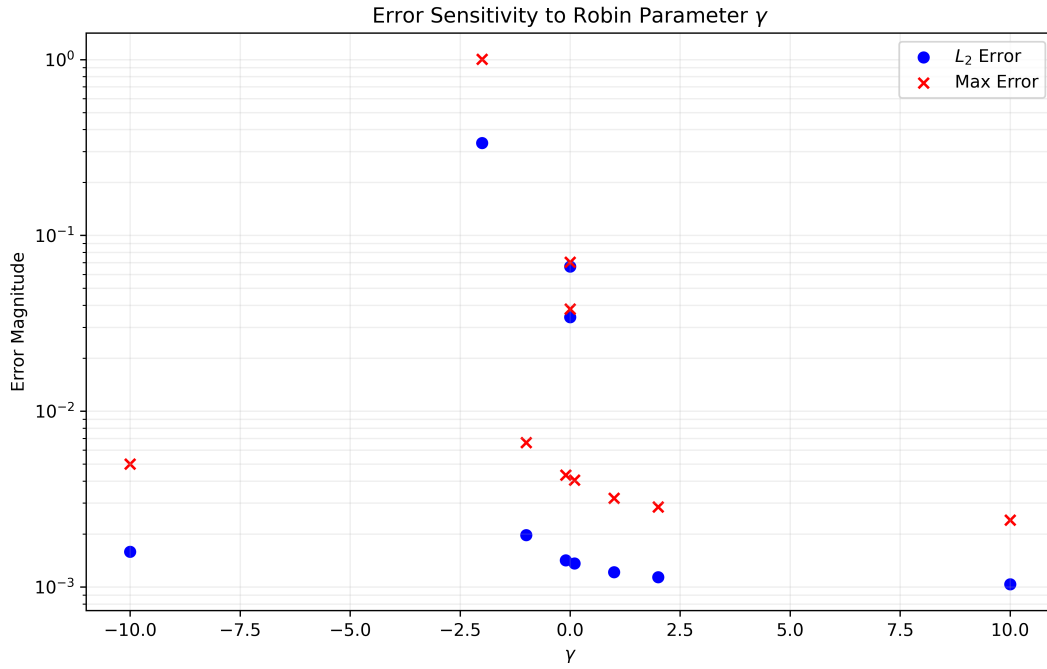


Figure 2: Error Study with various γ

Results [!h]

As expected, for a given $|\gamma|$, the positive γ has a lower L_2 error than the negative γ . This is expected from our previous discussion in part c. We also see the error increases as γ gets close to zero, which is expected as a gamma of $\pm 10^{-14}$ is almost zero as far as the computer is concerned. We also notice the positive gammas follow a trend where the higher gamma, the lower the L_2 error. However, the negative gammas appear unstable with the error seemingly following no pattern as the absolute value of gamma increases.

Table 1: Errors for varying γ

γ	L_2 Error	Max Error
1.0e-14	3.4287e-02	3.8121e-02
-1.0e-14	6.6471e-02	7.0319e-02
1.0e-01	1.3611e-03	4.0546e-03
-1.0e-01	1.4162e-03	4.3296e-03
1.0e+00	1.2134e-03	3.2018e-03
-1.0e+00	1.9724e-03	6.6243e-03
2.0e+00	1.1374e-03	2.8531e-03
-2.0e+00	3.3508e-01	1.0047e+00
1.0e+01	1.0360e-03	2.4025e-03
-1.0e+01	1.5854e-03	5.0005e-03

4 Biharmonic problem

4.A Weak Form

1. Start with the following

$$\int_{\Omega} d\mu \Delta^2 uv = \int_{\Omega} d\mu f v \quad (72)$$

2. GDT on the LHS

$$\int_{\Omega} d\mu \Delta^2 uv = - \int_{\Omega} d\mu \nabla(\Delta u) \nabla v + \int_{\Gamma} dS \frac{\partial \Delta u}{\partial n} v \quad (73)$$

3. We see the following term is zero because of the boundary conditions on v .

$$\int_{\Gamma} dS \frac{\partial \Delta u}{\partial n} v = 0, \quad v = 0 \in \Gamma \quad (74)$$

4. Apply GDT again

$$- \int_{\Omega} d\mu \nabla(\Delta u) \nabla v = \int_{\Omega} d\mu \Delta u \Delta v - \int_{\Gamma} dS \Delta u \frac{\partial v}{\partial n} \quad (75)$$

5. We see the following term is zero because of the boundary conditions on v_n

$$\int_{\Gamma} dS \Delta u \frac{\partial v}{\partial n} = 0, \quad \frac{\partial v}{\partial n} = 0 \in \Gamma \quad (76)$$

6. We stop applying GDT because we know nothing about $\Delta v \in \Gamma$.

$$\int_{\Omega} d\mu \Delta u \Delta v = \int_{\Omega} d\mu f v \quad (77)$$

$$a(u, v) = \int_{\Omega} d\mu \Delta u \Delta v \quad (78)$$

$$L(v) = \int_{\Omega} d\mu f v \quad (79)$$

4.B Symmetry, Continuity, Coersivity

Symmetry of bilinear form

1. We know the following

$$\Delta u \Delta v = \Delta v \Delta u \quad (80)$$

2. Therefore,

$$a(u, v) = \int_{\Omega} d\mu \Delta u \Delta v = \int_{\Omega} d\mu \Delta v \Delta u = a(v, u) \quad (81)$$

3. Bilinear form is symmetric

Continuity of bilinear form

1. Expand bilinear form and apply CS

$$|a(u, v)| = \left| \int_{\Omega} d\mu \Delta u \Delta v \right| \leq \|\Delta u\|_{L^2} \|\Delta v\|_{L^2} \quad (82)$$

2. Expand Sobolev norm

$$\|u\|_{H^2}^2 = \|u\|_{L^2}^2 + \|\nabla u\|_{L^2}^2 + \|\Delta u\|_{L^2}^2 \quad (83)$$

$$\|u\|_{H^2}^2 \geq \|\Delta u\|_{L^2}^2 \quad (84)$$

3. Apply above relationship

$$|a(u, v)| \leq \|\Delta u\|_{L^2} \|\Delta v\|_{L^2} \leq \|u\|_{H^2}^2 \|v\|_{H^2}^2 \quad (85)$$

4. See bilinear form is continuous with $C = 1$

$$|a(u, v)| \leq \|u\|_{H^2}^2 \|v\|_{H^2}^2 \quad (86)$$

Coersivity of bilinear form

1. Begin by expanding bilinear form

$$a(v, v) = \int_{\Omega} d\mu (\Delta v)^2 = \|\Delta v\|_{L^2}^2 \quad (87)$$

2. From 2b, we have the following

$$\|v\|_{H^2}^2 \leq C \int_{\Omega} d\mu |\Delta v|^2 \quad (88)$$

3. Combine the two expressions

$$a(v, v) = \int_{\Omega} d\mu |\Delta v|^2 \geq \alpha \|v\|_{H^2}^2 \quad (89)$$

4. We see, with $\alpha = 1/C$, the bilinear form is coercive

$$a(v, v) \geq \alpha \|v\|_{H2}^2 \quad (90)$$

Continuity of Linear form

1. Expand linear form using CS then apply the norm relationship we've used 100 times so far

$$|L(v)| = \left| \int_{\Omega} d\mu f v \right| \leq \|f\|_{L2} \|v\|_{L2} \leq \|f\|_{L2} \|v\|_{H2} \quad (91)$$

2. As long as f is square integrable over the domain, the L2 norm of f will be a constant. Therefore, we see the linear form is continuous

$$|L(v)| \leq \|f\|_{L2} \|v\|_{H2} = C \|v\|_{H2} \quad (92)$$

4.C FEniCSx

Gemini Prompt Finally, can you write the code for part c of this problem. Ensure you do the following:

1. Implement a Ciarlet-Raviart mixed formulation to decompose the fourth-order biharmonic equation into two coupled second-order PDEs.
2. Use `basix.ufl.element` and `basix.ufl.mixed_element` to define the function spaces according to the modern FEniCSx API.
3. Configure the `LinearProblem` with a `petsc_options_prefix` and the MUMPS direct solver to handle the indefinite saddle-point system.
4. Explicitly compute mesh connectivity using `domain.topology.create_connectivity` to allow the identification of boundary facets.
5. Apply `comm.allreduce` to local error scalars to ensure the L2 and Max errors are calculated globally across all MPI ranks.
6. Wrap all printing and plotting functions in a `rank == 0` conditional block to prevent redundant or conflicting outputs in parallel.
7. Use `V.sub(0).collapse()` to isolate the displacement subspace for the accurate application of Dirichlet boundary conditions.

Results [!h]

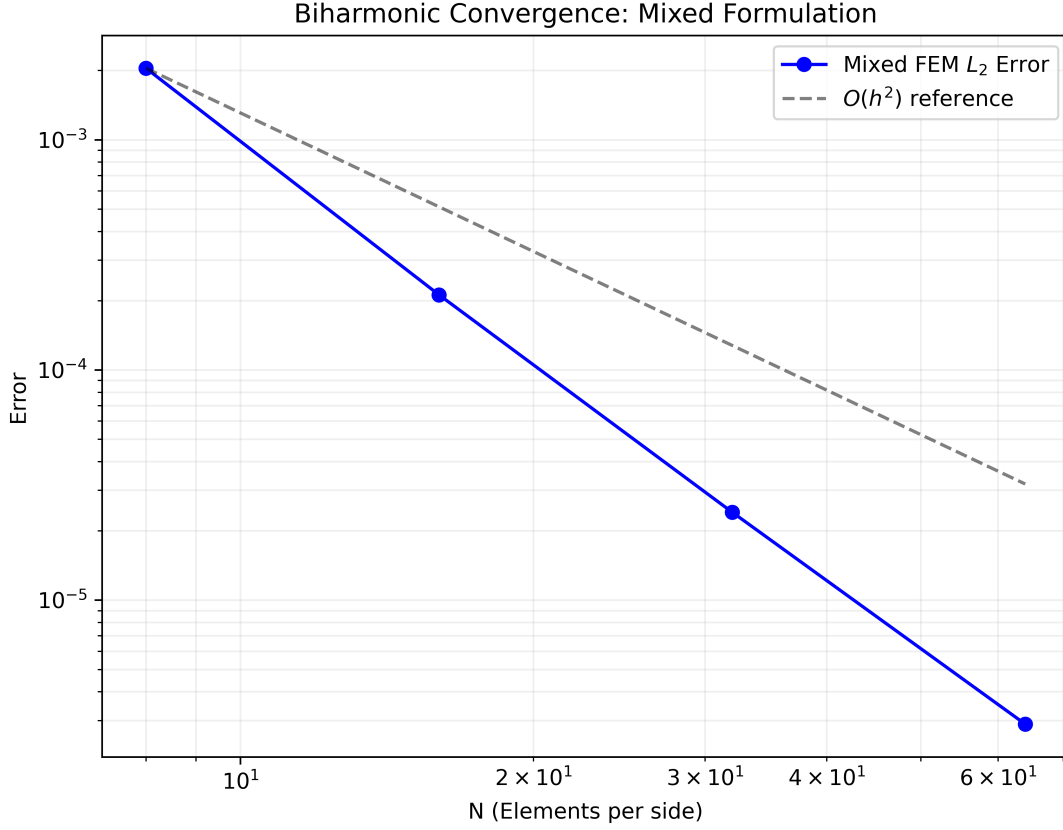


Figure 3: Biharmonic L_2 error convergence study as a function of elements per side

The mixed formulation is a method where we introduce an intermediate variable equal to $-\Delta u$. With this intermediate variable, we can split the fourth-order problem into two second-order problems. We split the problem into two second-order problems because, when we use Lagrange polynomials, the function is continuous, but the gradient can jump across elements. If the gradient can just across elements, the Laplacian becomes a delta function wherever you have the derivative discontinuity. If you then integrate the Laplacian when it is a delta function, the value goes to infinity. However, we can use the mixed formulation to remove the need to calculate the Laplacian. Since we do not need the Laplacian, we can use the Lagrange polynomials like always. We cannot apply the weak form from part a directly because it requires us to calculate the Laplacian of u, v – this causes our solution to freak out (this should prove this is my language because a robot like Gemini would never use the term "freak out").